Enhancing Urban Traffic Control with Priority-Aware Multi-Objective Reinforcement Learning

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***Abstract*—Efficient urban traffic control is crucial for enhancing public safety, reducing travel times, and ensuring timely responses from emergency vehicles, including ambulances and fire engines. While reinforcement learning (RL) has shown promise in adaptive traffic management, existing approaches often lack the flexibility to dynamically prioritize emergency vehicles. In this paper, we propose MOP-MADDPG, a Multi-Objective Prioritized extension of the Multi-Agent Deep Deterministic Policy Gradient framework, which enables decentralized traffic signal agents to optimize for both regular and emergency scenarios adaptively. Our method introduces context-aware learning and dynamically reweighted reward functions based on vehicle priority, ensuring emergency passage while maintaining overall traffic efficiency. We further perform hyperparameter tuning to optimize reward coefficients for robust performance. Experiments conducted on a real-world traffic network from Hanoi using the SUMO simulator demonstrate that MOP-MADDPG reduces waiting time by 75% and ensures a travel speed for emergency vehicles, while maintaining competitive performance for regular traffic.**

***Index Terms*—reinforcement learning, Q-learning, traffic management, multi-level agent, multi-objective reward**

I. INTRODUCTION

Optimizing urban traffic networks while accounting for emergency vehicles, such as ambulances and fire engines, presents a significant challenge. These scenarios introduce stringent time constraints and dynamic priorities, transforming the traffic signal control problem into a combinatorially complex task. In fact, balancing the efficient movement of regular traffic alongside the timely clearance of emergency vehicles is widely recognized as an NP-hard problem [1], [2]. Traditional rule-based traffic systems often fail to adapt quickly enough to such rapidly changing conditions, resulting in suboptimal traffic flow and delayed emergency responses.

RL has emerged as a promising solution for adaptive traffic signal control [3], enabling agents to learn optimal strategies directly from interaction with the environment. However, existing approaches often treat all vehicles equally or rely on fixed preemption rules, lacking the flexibility to adjust priorities in real-time when emergency scenarios arise dynamically.

To address these limitations, we propose a novel approach based on an enhanced Multi-Agent Deep Deterministic Policy Gradient (MADDPG) framework [4], augmented with context awareness and dynamic reward shaping. Our method, called MOP-MADDPG (Multi-Objective Prioritized MADDPG), is designed to adapt its behavior based on the presence and location of emergency vehicles while maintaining overall traffic efficiency [5]. Specifically, we introduce a dynamic reward function that is reweighted depending on traffic conditions and vehicle priority, ensuring that emergency vehicles receive immediate passage through intersections without severely penalizing the performance of regular traffic flow [6].

Our main contributions to this paper are as follows: Firstly, we incorporate contextual awareness into the learning framework to distinguish between normal and emergency traffic scenarios, allowing agents to adapt their strategies accordingly. Secondly, we develop a multi-objective reward formulation that adjusts weights based on the presence and urgency of emergency vehicles, integrating multiple traffic performance metrics, such as queue length, waiting time, and vehicle speed. Finally, we conduct extensive hyperparameter tuning to identify optimal reward coefficient configurations, ensuring balanced and resilient performance under diverse traffic densities and emergency conditions.

We evaluate the proposed framework on a real-world traffic network extracted from a Hanoi map section using the SUMO simulator. Experimental results demonstrate that our MOP-MADDPG method significantly reduces 30-50% waiting times and improves average travel speeds for both regular and emergency vehicles compared to baseline methods.

The rest of our paper is as follows: Section II summarizes existing solutions in the literature. Section III provides background knowledge on reinforcement learning. The research problem is modeled in Section IV. Our proposal solution is given in Section V. Section VI provides experiment results to validate our proposal. Finally, we conclude in Section VII.

II. Related works

The literature implies that optimizing traffic flow problems for emergency vehicles requires intelligent control strategies that minimize waiting times while prioritizing EVs. Reinforcement learning has been widely applied to this problem, yielding diverse approaches. We survey a series of scientific papers that apply RL to traffic management with EV priority, comparing their RL techniques, unique features, and pros/cons. One may classify them into Deep Q-learning methods, Actor-Critic (policy gradient) methods, and Multi-Agent RL coordination.

Deep Q-learning techniques [3], [7], [8] use neural networks to approximate the Q-value function, enabling an agent to learn optimal actions for traffic signals. These approaches treat traffic signal timing as a discrete action selection problem, where the goal is to minimize waiting times (or queue lengths) with heavy penalties for delaying emergency vehicles. Despite huge improvements([7] reports a 40% decrease in emergency vehicle delay under multiple-EV scenarios), the pure Q-learning approach still suffers from some limitations due to its discretization nature.

Actor-critic RL methods (including policy gradient algorithms like Advantage Actor-Critic and Proximal Policy Optimization) have been applied to traffic problems to handle continuous or large action spaces and provide more stable convergence [9]–[11]. In the context of emergency vehicle priority, actor-critic approaches often shine when the action involves timing durations or continuous maneuvers (like vehicle acceleration/lane-changing) or when combining decisions in a parameterized way. In those works, actor-critic methods are chosen for scenarios requiring either continuous actions (timing or vehicle control) or stable multi-agent learning. Besides making decisions, coordinating multiple agents in optimizing traffic flow problems is also crucial due to the correlation between multiple traffic intersections.

Multi-agent reinforcement learning (MARL) is central to managing traffic networks where multiple intersections or vehicles must work in tandem. Several of the above papers already incorporate MARL elements (EMVLight [9], RL-CMAS [10], and EVPRT [12] all involve multiple agents). Multi-agent RL in traffic EV priority is about balancing cooperation and complexity. Methods like EVPRT and EMVLight introduced ways for infrastructure agents to cooperate (via shared policies or communication networks). At the same time, [6], [13] focus on vehicle agents cooperating or even multiple EVs cooperating. A common thread is the use of centralized training, giving the MARL system a bird's-eye view during learning, to achieve coordination that decentralized methods (each agent selfishly optimizing) would miss.

Techniques such as value decomposition (Q-LSTM) [13] and graph neural networks (GAT in EVPRT) [12] are employed to handle the combinatorial complexity of multi-agent interactions by imposing structure (factorizing the value or exploiting road network topology). These innovations make the RL more sample-efficient and the learned policies more effective in a multi-agent context.

III. Preliminary

A. Preliminary

In an RL problem, $D = \{(s_t, a_t, r_t, s_{t+1})\}$ is a set of data tuple consisting of a current state s_t, a taken action a_t, a reward function r_t, and a transition must be defined s_{t+1}. At each time step t, the agent is in a state $s_t \in S$ and selects an action $a_t \in A$. The environment evolves by transitioning the agent to a new state s_{t+1} according to the transition function: $T(s_t, a_t)$ Simultaneously, a reward r_t is given to the agent that is expressed by the reward function: $R(s_t, a_t, s_{t+1})$

The agent needs to respect a policy $\pi(s)$, which defines the probability of selecting one of the available actions. The objective is to determine the optimal policy π that optimizes the accumulated reward R. Learning an optimal policy is to estimate the state-action value function $Q^\pi(s, a)$. This function is defined as: $Q^\pi(s, a) = \mathbb{E}\left[R \mid s_t = s, a_t = a\right]$

B. Multi-Agent Deep Deterministic Policy Gradient - MADDPG

MADDPG is a multi-agent reinforcement learning algorithm designed to address the challenges of coordination and stability in environments involving multiple interacting agents. During training, each agent, a Q-network, maintains its own actor–critic architecture. The actor learns a deterministic policy based on local observations. At the same time, the critic is trained with access to global information, including the observations and actions of all agents in the environment.

The reward for agent ag_i at time step t is given by: $r_i^t = R_i(s^t, a_1^t, a_2^t, \ldots, a_n^t)$ where a_i^t is the action taken by the agent ag_i at time t. Each agent in MADDPG uses a centralized critic that considers the global state and actions of all agents $Q_i(s, a_1^t, a_2^t, \ldots, a_n^t)$. The critic is updated using the Bellman equation: $y_i = r_i + \gamma Q_i(s^{t+1}, a_1^{t+1}, a_2^{t+1}, \ldots, a_n^{t+1})$. The actor policy μ_i for agent ag_i is updated using the deterministic policy gradient:

$$\nabla_{\theta_i} J(\theta_i) = \mathbb{E}_{s,a\sim D}\left[\nabla_{\theta_i}\mu_i(o_i)\nabla_{a_i}Q_i(s, a_1, \ldots, a_n)\right]$$

IV. Problem statements

In this context, we address the RL problem of traffic management in a realistic urban scenario. A traffic network is modeled using a real-world map comprising n signalized intersections. Each intersection is managed by a corresponding agent ag_i where $i = 1, 2, ..., n$. These agents interact with the environment by receiving observations in the form of feature vectors that capture the dynamic aspects of nearby vehicles and traffic signals.

A. States

Each agent ag_i works on the data tuple $D_i = \{(s_t, a_t, r_t, s_{t+1})\}$. At each time step t, the environment provides a state s_t, which includes the following components:

Each intersection may support up to several-way, in each way, there are some possible lanes. So, each intersection is totally described by p lanes. Fig. 1 captured from SUMO [1] shows 20 different lanes in an intersection.

For each lane in the way, we define traffic data as $d_i = \{v_m^i, v_h^i\}$ where $i = 1, \ldots, p$ and v_m^i, v_h^i denote the number of moving and halting vehicles, respectively. Fig. 2, captured from SUMO, illustrates an example of these vehicle types in an intersection. Typically, moving vehicles are observed on green-light lanes; otherwise, halting vehicles accumulate before red lights. A state s_t encapsulates the full distribution of both moving and stopping vehicles across all lanes in the road network.

Additionally, the state at time t should also contain the position and waiting time of any emergency vehicle present in the intersection. The position is represented by a one-hot vector of length p, where value 1 indicates the exact location of the emergency vehicle.

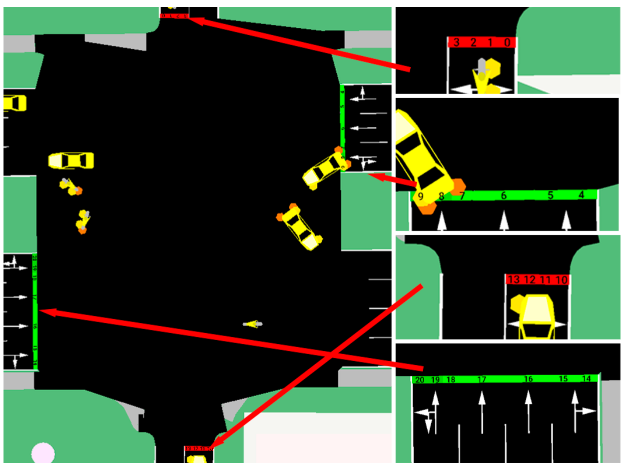


Fig. 1: Moving direction samples in an intersection.

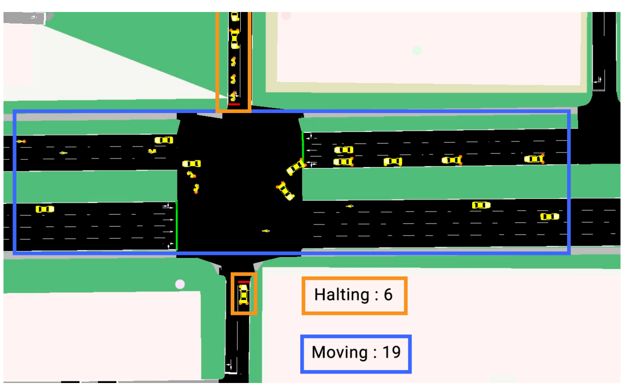


Fig. 2: Sample for moving and halting vehicles in an intersection.

B. Actions

At each step, an agent ag_i selects an action a_t to transition the system from state s_t to the next state s_{t+1}. In this setup, an action corresponds to toggling the traffic light phase—either switching from red to green or vice versa. Each phase is held for a vary duration to allow vehicles to proceed. For simplicity in implementation, traffic light phases are encoded in binary form, with green = 1 and red = 0.

Each intersection is equipped with q traffic lights, represented as a binary vector of length q, ordered as [North, East, South, West]. For example, in a standard four-way intersection, the vector [1, 0, 1, 0] indicates that the north–south direction has green lights, while the east–west direction has red lights.

C. Regular Rewards

The reward function in our framework is carefully formulated to enable each agent ag_i to make effective traffic signal switches for both regular and emergency traffic conditions. Decentralized agents are expected to prioritize the passage of these vehicles over others to ensure unobstructed movement.

Under regular traffic conditions, rather than relying on a single objective, the reward integrates multiple traffic performance metrics [14], [15], enabling the agent to make decisions that contribute to system-wide optimization. The following criteria are selected to evaluate the reward function:

- Waiting time f_{wt}: The total cumulative waiting time of vehicles approaching or halting at the intersection is minimized. By reducing waiting time, the system aims to enhance overall traffic flow efficiency.
- Queue length f_{ql}: The agent attempts to minimize the number of vehicles queued at each lane. A shorter queue length generally reflects smoother traffic movement and reduced congestion. The change δf_{ql} in f_{ql} between time steps t and $t+1$ should be considered in the reward function; however, only the remaining queue length is currently incorporated.
- Vehicle speed f_{sp}: The system encourages higher average speeds within safe operational limits. Low speeds may indicate congestion, while consistently high speeds (when appropriate) reflect effective signal timing decisions.

Thus, under normal conditions, the reward function is affected by the set of performance factors $fg_i = \{f_{wt}, f_{ql}, \delta f_{ql}, f_{sp}\}$ where i = 1, ..., 4.

D. Emergency Rewards

In emergency scenarios, the reward function must be adaptively adjusted to account for the presence and priority of emergency vehicles with the following factors:

- Emergency queue length $\mathfrak{f}_{ql}$: Similar to the queue length in regular traffic, we propose a reward mechanism that incentivizes changes in queue length $\delta\mathfrak{f}_{ql}$ while imposing penalties when queues remain persistent.
- Waiting time $\mathfrak{f}_{wt}$: The reward function penalizes delays encountered by emergency vehicles. Minimizing their waiting time becomes a primary objective.
- Emergency vehicle speed $\mathfrak{f}_{sp}$: The high speed of emergency vehicles is favored in the reward. The agent should emphasize maintaining or increasing this speed by adjusting signal phases.

[1] https://sumo.dlr.de/docs/index.html

Similarly, in emergency conditions, the reward function is affected by the set of performance factors $\mathfrak{f}e_i = \{\mathfrak{f}_{wt}, \mathfrak{f}_{ql}, \delta\mathfrak{f}_{ql}, \mathfrak{f}_{sp}\}$ where i = 1, ..., 4. This emergency-aware reward design ensures that agents can prioritize critical traffic in real time while still retaining the ability to return to standard optimization goals once the emergency scenario subsides.

V. Proposed Method

Our approach, MOP-MADDPG, builds upon the MADDPG algorithm. However, our method introduces a priority-aware extension specifically designed to handle emergency traffic scenarios. The key point is to adjust the reward function based on the context of involving emergency vehicles.

A. Agents

An agent ag_i is described by: a local policy $\pi(s)$ to choose optimal actions at a given state; and a centralized critic $Q^{\pi}(s, a_1^t, a_2^t, \ldots, a_n^t$ determined by the actions of all n agents. During training, the centralized critic uses global information to stabilize learning. However, the agents operate independently at test time while still benefiting from global knowledge during learning. A feed-forward neural network of two 64-unit layers is implemented to estimate the Q-value for each agent's action.

B. Reward Design

In normal conditions, the reward function focuses on general traffic optimization metrics, such as waiting time, queue length, and average speed. However, in emergency scenarios, the reward is reweighted to prioritize emergency vehicle throughput. These factors are emphasized during emergencies, allowing the agent to adapt its behavior in real time. Each agent ag_i receives a reward defined by the following function:

$$r_i = \sum_{j=1}^{4} \alpha_j \cdot fg_j + \sum_{j=1}^{4} \gamma_j \cdot fe_j \tag{1}$$

where α_j and γ_j correspond to the weighting coefficients for each component of the reward factors fg_j and fe_j, respectively. These weights are treated as hyperparameters and are carefully tuned to optimize the performance of the reward function in our experiments. This ensures that emergency vehicles are given preferential treatment in signal phase decisions, which can significantly reduce their travel time and improve overall response effectiveness.

C. Bellman Equation Update

In the proposed method, each agent maintains an individual actor and critic network. The Bellman equation is used to compute the target Q-value during training, which forms the basis for updating the critic network through the minimization of the temporal-difference error. This allows the agent to learn the expected return of its actions in the presence of other learning agents.

For each agent ag_i, the Bellman target is defined as:

$$y_i = r_i + \gamma(1-d) \cdot Q_i^{\text{target}}(s^{t+1}, a_1^{t+1}, a_2^{t+1}, \ldots, a_n^{t+1})$$

where:

- r_i is the immediate reward received by agent ag_i,
- $\gamma \in [0, 1)$ is the discount factor,
- $d \in \{0, 1\}$ is the terminal flag indicating the end of an episode,
- Q_i^{target} is the target critic network for agent ag_i,
- s^{t+1} is the next global state after taking action at time t,
- a_j^{t+1} is the next action for agent ag_j, computed by the corresponding target actor network.

The current Q-value is obtained from the critic network as $Q_i(s^t, a_1^t, a_2^t, \ldots, a_n^t)$, where s^t is the current global state and a_j^t denotes the action of agent j at time step t. The critic loss function is defined as the mean squared error between the predicted Q-value and the Bellman target:

$$\mathcal{L}_i^{\text{critic}} = \frac{1}{B} \sum_{k=1}^{B} \left(Q_i(s_k^t, a_k^t) - y_i^k \right)^2$$

where B is the batch size, and the summation is taken over a minibatch of transitions sampled from the replay buffer.

The model employs a centralized training framework with decentralized execution by conditioning each agent's critic on the actions of all agents. This allows the learning process to remain stable in the presence of multiple agents while supporting independent action selection during test time.

D. Exploration Strategy

We employ a ϵ-greedy strategy, a common and effective approach in reinforcement learning for balancing the trade-off between exploration and exploitation. Given a state s_t, an agent ag_i selects an action based on a probabilistic decision given by a Q-network as following

- **Exploration:** With probability ϵ, the agent selects a random action from the action set. This phase is crucial during the early stages of training, where the policy is underdeveloped and broad exploration is needed to discover informative state-action transitions.
- **Exploitation:** Otherwise, the agent selects the action that maximizes its own critic's estimate of the Q-value, conditioned on the joint state and actions of all agents.

$$a_i^t = \arg\max_{a_i} Q_i(s^t, a_1^t, \ldots, a_i, \ldots, a_N^t)$$

VI. Experiments

We conduct a variety of experimental scenarios to evaluate and compare the effectiveness of the proposed method against baseline approaches. A specific area of Hanoi is chosen for detailed analysis, using accurate urban road information. The corresponding map is shown in Fig. 3, where five key intersections are highlighted, each representing an individual agent.

A. Models for Comparisons

In Table I, the State column represents the length of the state vector used during training. A state with full-lanes contains all moving and halting vehicles of every lane in the network. For MADDPG 2, local lane refers to training without

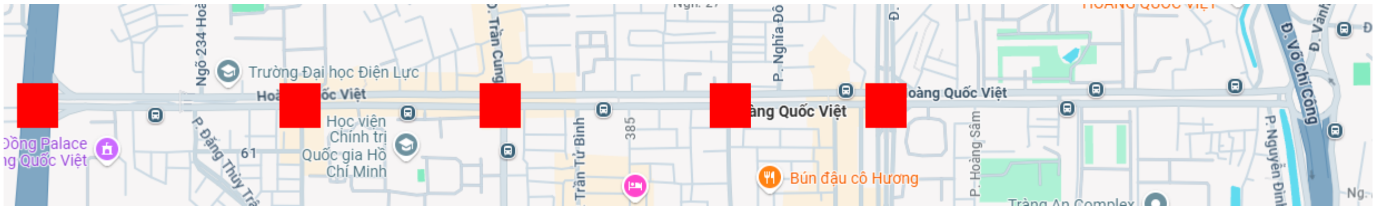


Fig. 3: Actual map of Hoang Quoc Viet street in Hanoi with 5 selected light traffic (extracted from Google Map).

TABLE I: Comparison of Reinforcement Learning Models

Model	State	Reward	Emergency
MOP-MADDPG	Full lanes	Multi-objective	Yes
MADDPG 1	Full lanes	Multi-objective	No
MADDPG 2	Local lanes	Multi-objective	Yes
MADDPG 3	2nd-order lanes	Multi-objective	Yes

incorporating the state from neighboring lanes. In contrast, second-order lanes of MADDPG 3 indicate that the agent considers two levels of neighboring lanes in its input. The Emergency column describes whether the reward function is integrated with emergency factors.

B. Environment Settings

We need to select several environment parameters for SUMO as follows:

- Experiments are trained a total number of 1,720 vehicles, including 4 emergency vehicles. The random route algorithm of SUMO handles their appearance. We exploit the exploration step for the scenario of 2,580 vehicles, including 6 emergency vehicles.
- Each vehicle is characterized by its type (car or motorbike) and speed. The speeds are randomly assigned within the range [0, 50] km/h for cars, [0, 40] km/h for motorbikes, and [0, 60] km/h for emergency vehicles.
- The duration of each light phase is set to 10 seconds for regular traffic conditions. In contrast, for emergency scenarios, it is evaluated and potentially switched every 1 second.
- Each model is trained and evaluated over 100 epochs to ensure performance consistency and convergence.

C. Hypertuning

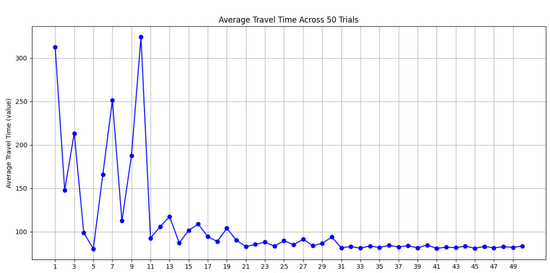


Fig. 4: Average waiting time of 1720 regular vehicles excluding emergency vehicles in 50 combination trials.

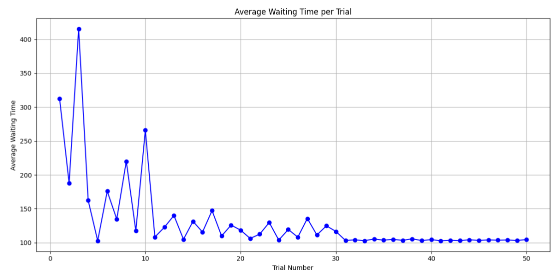


Fig. 5: Average waiting time of 1716 regular vehicles and 4 emergency vehicles in 50 combination trials.

With the previous setting, we examine the coefficient factors α and γ in Equation 1. Since vehicles require a certain amount of time to pass through intersections, the impact of queue length on traffic performance is more significant compared to factors such as waiting time and speed. Consequently, the corresponding reward weights for queue length and its variation should be assigned higher values, whereas lower weights are chosen for the remaining factors. The objective in this step is to identify the optimal combination (trial) of coefficient factors within a specified range that minimizes the average travel time, thereby achieving the best overall traffic performance.

- Regular factors: $\alpha = \{\alpha_{f_{wt}} = [0.0005, 0.15], \alpha_{f_{ql}} = [-0.6, -0.05], \alpha_{\delta f_{ql}} = [0.05, 1.5], \alpha_{f_{sp}} = [0.01, 0.6]\}$
- Emergency factors: $\gamma = \{\gamma_{\mathfrak{f}_{wt}} = [-15.0, -1.0], \gamma_{\mathfrak{f}_{ql}} = [-0.15, -0.01], \gamma_{\delta \mathfrak{f}_{ql}} = [1.0, 15.0], \gamma_{\mathfrak{f}_{sp}} = [-6.0, 0.0]\}$

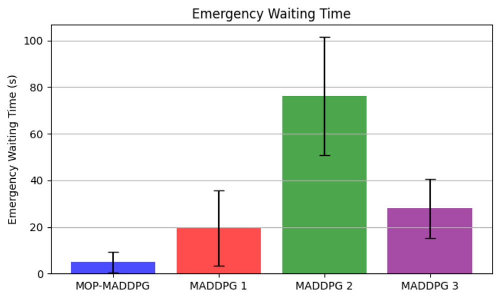
We execute 50 uniformly distributed trials within the defined range to find the best trials, as shown in Fig. 4 and 5. In Fig. 4, Trial 5 achieves the best performance with the coefficient configuration $\{\alpha_{f_{wt}} = 0.01, \alpha_{f_{ql}} = -0.2, \alpha_{\delta f_{ql}} = 0.5, \alpha_{f_{sp}} = 0.2\}$. In Fig. 5, the optimal result is obtained with the coefficient set $= \{\gamma_{\mathfrak{f}_{wt}} = -5.0, \gamma_{\mathfrak{f}_{ql}} = -0.05, \gamma_{\delta \mathfrak{f}_{ql}} = 5, \gamma_{\mathfrak{f}_{sp}} = -2.0\}$. These optimized combinations of coefficients are adopted in our experiments to configure the reward function for the learning framework.

D. Evaluation

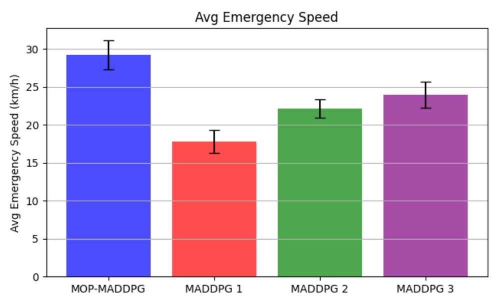
Figure 6 presents a comparison of the average waiting times for emergency scenarios across the models listed in Table I. The average results in this table are based on 20 simulation runs involving 2,580 vehicle. In Figure 6a, the bars represent the mean waiting time in seconds, while the error bars indicate the standard deviation for priority vehicles. The

TABLE II: Performance comparison of different models.

Model	**Emergency Vehicle**			**All Vehicles**	
(Mean ± std)	**Waiting Time (s)**	**Travel Time (s)**	**Speed (km/h)**	**Travel Time (s)**	**Speed (km/h)**
MOP-MADDPG	**4.94 ± 4.38**	**113.37 ± 5.78**	**29.20 ± 1.94**	94.54 ± 1.54	29.20 ± 0.14
MADDPG 1	19.57 ± 16.13	129.78 ± 18.36	17.78 ± 1.51	98.77 ± 2.88	28.66 ± 0.36
MADDPG 2	76.13 ± 25.45	188.01 ± 26.04	22.14 ± 1.19	95.41 ± 1.96	**30.00 ± 0.22**
MADDPG 3	27.92 ± 12.58	140.54 ± 12.75	23.94 ± 1.70	**92.45 ± 0.91**	29.27 ± 0.11



(a) Performance on average waiting time of 6 emergency vehicles.



(b) Performance on average speed time of 6 emergency vehicles.

Fig. 6: Comparison in different performance measures.

MOP-MADDPG model achieves the lowest waiting time at approximately 4.94 seconds , a 75% reduction in delay for emergency vehicles, but keeps good speed around 29.2km/h for six emergency vehicles as in Fig. 6b. MOP-MADDPG is still able to maintain comparable travel times and speeds for all vehicles within 3–5% of those achieved by the other models..

Moreover, Table. II demonstrates the good performance of the proposed MOP-MADDPG method over the baseline MADDPG variants according to multiple metrics. Most notably, MOP-MADDPG achieves the best scores for the emergency scenario. This indicates that MOP-MADDPG responds more promptly to critical scenarios.

VII. Conclusion

In this work, we proposed MOP-MADDPG, a novel Multi-Objective Prioritized Multi-Agent Deep Deterministic Policy Gradient framework for managing traffic environments. Our approach integrates multi-objective optimization with the MADDPG algorithm, enabling agents to balance general traffic flow while prioritizing emergency vehicles efficiently. Through comprehensive experiments and comparisons with multiple MADDPG variants, we demonstrated that MOP-MADDPG significantly outperforms baseline methods in terms of emergency response efficiency, achieving the lowest emergency waiting and travel times. Furthermore, it maintains competitive average travel and waiting times for all vehicles. Future work will explore scalability to larger networks with real-time data.

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